

Compound 1 shown above is represented as a line-angle structure. Which of the following condensed structures is a correct representation of Compound 1?

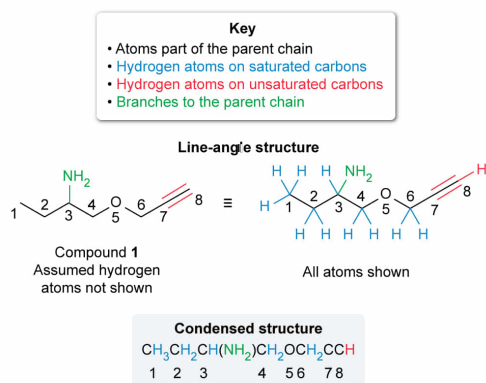
- ✓ ☒ A. $\text{CH}_3\text{CH}_2\text{CH}(\text{NH}_2)\text{CH}_2\text{OCH}_2\text{CCH}$ [92%]
☐ B. $\text{CH}_3\text{CH}_2\text{CH}(\text{NH}_2)\text{COCH}_2\text{CH}_2\text{CCH}$ [2%]
☐ C. $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}(\text{NH}_2)\text{OCH}_2\text{CCH}$ [1%]
☐ D. $\text{CH}_3\text{CH}_2\text{CH}(\text{NH}_2)\text{CH}_2\text{OCH}_2\text{CCH}_3$ [3%]

Correct

92% answered correctly

Explanation:

Converting line-angle structures to condensed structures



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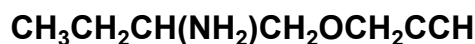
Organic molecules can be depicted in several ways. Condensed and line-angle structures are a shorthand way of representing molecular structure because these approaches do not show all atoms or bonds. **Line-angle structures** emphasize the carbon skeleton of a molecule. **Carbon atoms** are **implied** at the end of a line and wherever the line changes direction (ie, at each vertex), and enough hydrogen atoms are implied to satisfy the octet rule. Heteroatoms (eg, oxygen, nitrogen) and multiple bonds (ie, double and triple bonds) are always shown.

Condensed structures show **all atoms present** in a molecule but do not explicitly show all bonds. **Branch points** are indicated by **parentheses**, and **nonbonding electrons** and **multiple bonds** are **inferred** based on the **octet rule**.

In this question, the parent chain of Compound 1 contains eight atoms (seven carbons and one oxygen). Numbering the parent chain from left to right, a **primary amine** (NH_2) branches from carbon 3, which is indicated with parentheses after carbon 3 in the condensed structure. Carbon 3 has one assumed hydrogen in addition to the single bonds to nitrogen and two adjacent carbons. Carbons 4 and 6 are each bonded to one carbon and the **ether** oxygen;

consequently, these carbons each have two implied hydrogens to fulfill their octet. The triple bond at the end of the chain is between carbons 7 and 8. The octet for carbon 7 is full, given the single bond to carbon 6 and the triple bond to carbon 8. Carbon 8 has one assumed hydrogen atom in addition to the triple bond.

Putting all this information together, the condensed structure for Compound 1 is written as:



(Choice B) The structure $\text{CH}_3\text{CH}_2\text{CH}(\text{NH}_2)\text{COCH}_2\text{CH}_2\text{CCH}$ incorrectly indicates an additional carbon atom exists in the molecule. The CO portion indicates a *carbonyl* group instead of an ether.

(Choice C) The structure $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}(\text{NH}_2)\text{OCH}_2\text{CCH}$ incorrectly places the amine (NH_2) on carbon 4.

(Choice D) The structure $\text{CH}_3\text{CH}_2\text{CH}(\text{NH}_2)\text{CH}_2\text{OCH}_2\text{CCH}_3$ incorrectly has three hydrogens on carbon 8 instead of just one, which breaks the octet rule.

Educational objective:

Line-angle structures always depict heteroatoms and multiple bonds but imply the carbon and hydrogen atoms. Condensed structures show all atoms present, and the branch points are written in parentheses.

Time spent: 0 sec

Subject:
Organic Chemistry

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System:
Introduction to Organic Chemistry